

Differential and Integral Calculus 1M 104041

Midterm
November 23, 2024

Your ID Number: _____

Your Name: _____

Guidelines

- **Duration: 3 hours.** Use of calculators, personal dictionaries, electronic devices, reference materials, personal notes or any other extra material is not allowed.
- **In exercises 1-6,** choose the answer that you believe is correct according to the statements.
- **In exercises 7-8,** show all your work. Explain your solutions, quote theorems you are using. **No credit will be given for non-justified answers!**
Write clear and complete answers for each problem in the same page.

Let $f(x) = \sqrt{1 + \sqrt{e^x + 8}}$ and $g(x) = |x^3 + x^2|$.

1. **(10 points)** Choose the right answer:
 - a. f is increasing and g is neither even nor odd.
 - b. f is increasing and g is even.
 - c. f is decreasing and g is odd.
 - d. f^{-1} is increasing and g is odd.
 - e. f^{-1} is decreasing and g is even.

Let $f(x) = \sqrt{1 + \sqrt{e^x + 8}}$ and $g(x) = |x^3 + x^2|$.

2. **(10 points)** Choose the right answer:

- a. f is differentiable on $\mathbb{R}$ and g is differentiable on $\mathbb{R} \setminus \{-1\}$.
- b. f is differentiable on $\mathbb{R}$ and g is differentiable on $\mathbb{R} \setminus \{-1, 0\}$.
- c. f and g are differentiable on $\mathbb{R}$.
- d. $g'(-1) = 24f'(0)$.
- e. $g'(0) > f'(0)$.

Consider the function

$$f(x) = \begin{cases} \frac{\sin(\frac{\pi x}{2})}{x} & \text{if } x < 0, \\ \arctan\left(\ln\left(\frac{\cos^2(x)}{x}\right)\right) & \text{if } x > 0. \end{cases}$$

3. **(10 points)** Choose the right answer:

- a. The function has two horizontal asymptotes at $y = -\frac{\pi}{2}$ and $y = 0$.
- b. The function has two horizontal asymptotes at $y = \frac{\pi}{2}$ and $y = 0$.
- c. The function has a horizontal asymptote at $y = \frac{\pi}{2}$ and a vertical asymptote at $x = \frac{\pi}{2}$.
- d. The function has a horizontal asymptote at $y = -\frac{\pi}{2}$ and a vertical asymptote at $x = \frac{3\pi}{2}$.
- e. The function has a horizontal asymptote at $y = 0$ and a vertical asymptote at $x = \frac{\pi}{2}$.

Consider the function

$$f(x) = \begin{cases} \frac{\sin(\frac{\pi x}{2})}{x} & \text{if } x < 0, \\ \arctan\left(\ln\left(\frac{\cos^2(x)}{x}\right)\right) & \text{if } x > 0. \end{cases}$$

4. **(10 points)** Choose the right answer:

- a. f can be defined as a continuous function at $x = 0$.
- b. f is continuous on $(0, \infty)$.
- c. $\lim_{x \rightarrow 0^-} f(x) \neq \lim_{x \rightarrow 0^+} f(x)$.
- d. f is continuous at $x = \frac{\pi}{2}$.
- e. f is discontinuous only at $x = 0$ and at $x = \frac{3\pi}{2}$.

Consider the curve C defined as the set of points verifying the following equation

$$\cos(2\pi x + 3\pi y) = \arctan(xy^3) + x^2 - 2 - \frac{\pi}{4}.$$

5. **(10 points)** Choose the right answer:

- a. The tangent line to C at $(1,1)$ is parallel to the tangent line to the hyperbola $xy = 15$ at $(3,5)$.
- b. The tangent line to C at $(-1, -1)$ is parallel to the tangent line to the curve $h(x) = \sin\left(\frac{5}{3}x\right)$ at $\left(\frac{6\pi}{5}, h\left(\frac{6\pi}{5}\right)\right)$.
- c. Suppose that (x, y) and $(-x, -y)$ are in the curve; then, the tangent line at $(-x, -y)$ is perpendicular to the tangent line at (x, y) .
- d. The tangent line to C at $(1,1)$ is perpendicular to the line $3y = 5x + \pi$.
- e. The tangent line to C at $(1,1)$ is horizontal.

Let $f(x) = \frac{\cosh x + 1}{\cosh x - 2}$.

6. **(10 points)** Choose the right answer:

- a. f has two vertical asymptotes.
- b. The lines $x = 3 - \sqrt{2}$ and $x = 3 + \sqrt{2}$ are vertical asymptotes of f .
- c. f is increasing.
- d. There is an x in $\mathbb{R}$ such that $f(x) = 1$.
- e. f is one-to-one.

7. **(20 points)** Assuming that $\lim_{t \rightarrow 0} \frac{\sin(t)}{t} = 1$, prove by the definition with ϵ - δ that

$$\lim_{t \rightarrow 0} \left(\frac{\sin(t)}{t} - 1 \right)^2 = 0.$$

8. (20 points) Consider the function for $c \in \mathbb{R}$:

$$f(x) = \begin{cases} x^{2/3} & \text{if } x \leq 1, \\ \frac{x^3 - 5x^2 + 8x - 4c}{(x-1)} & \text{if } x > 1. \end{cases}$$

- a. Is there any c such that $f'(1)$ exists? Justify your answer using the definition of derivative.
- b. Find all $x \in \mathbb{R}$ such that $f(x)$ is differentiable.